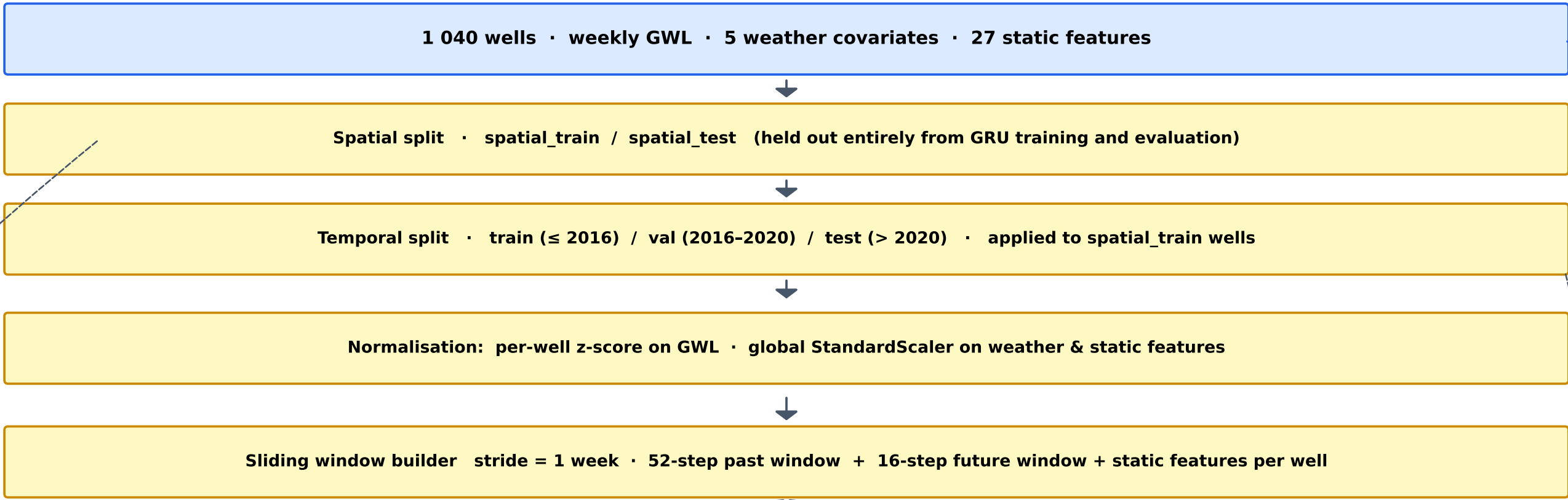


GRU + GP Decoupled Training — Groundwater Level Forecasting Architecture

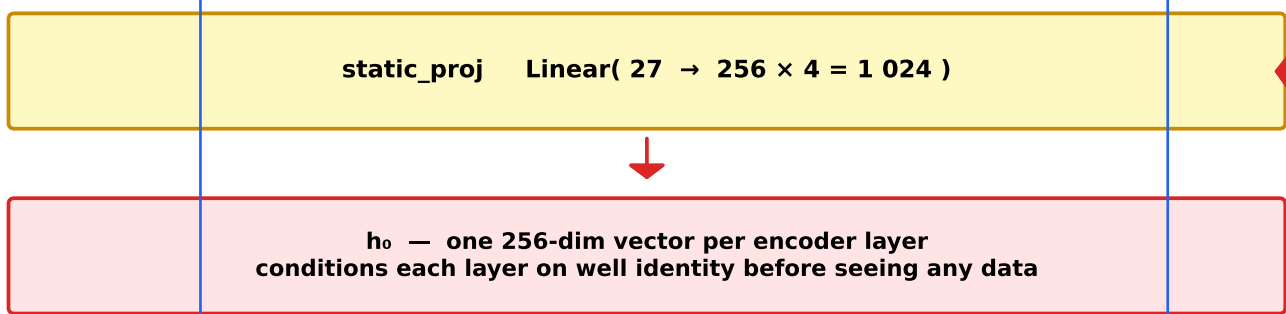
① Data Preparation



② Model Inputs (batch size B)

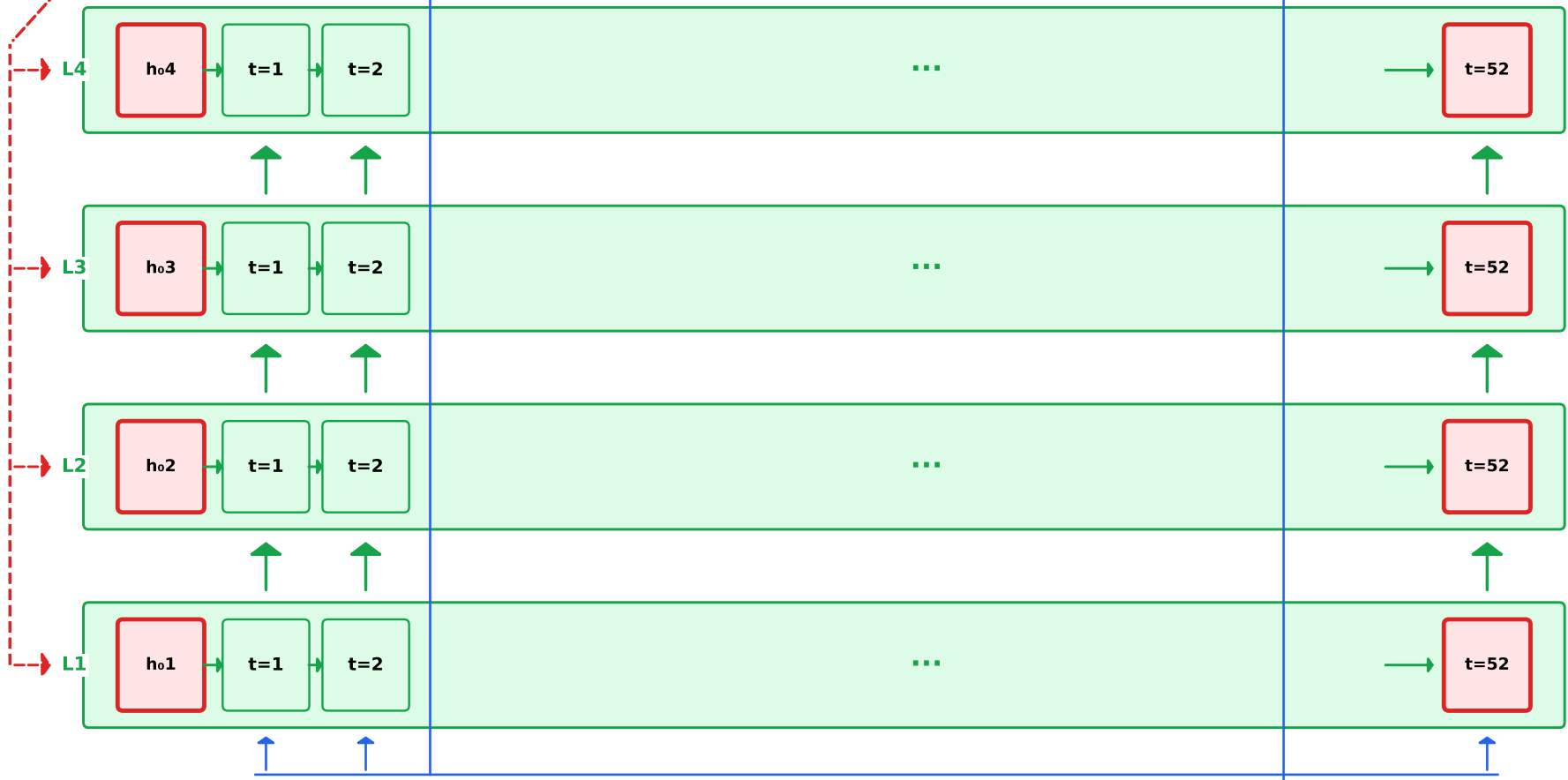


③ Static Projection → Encoder Initial State h_0



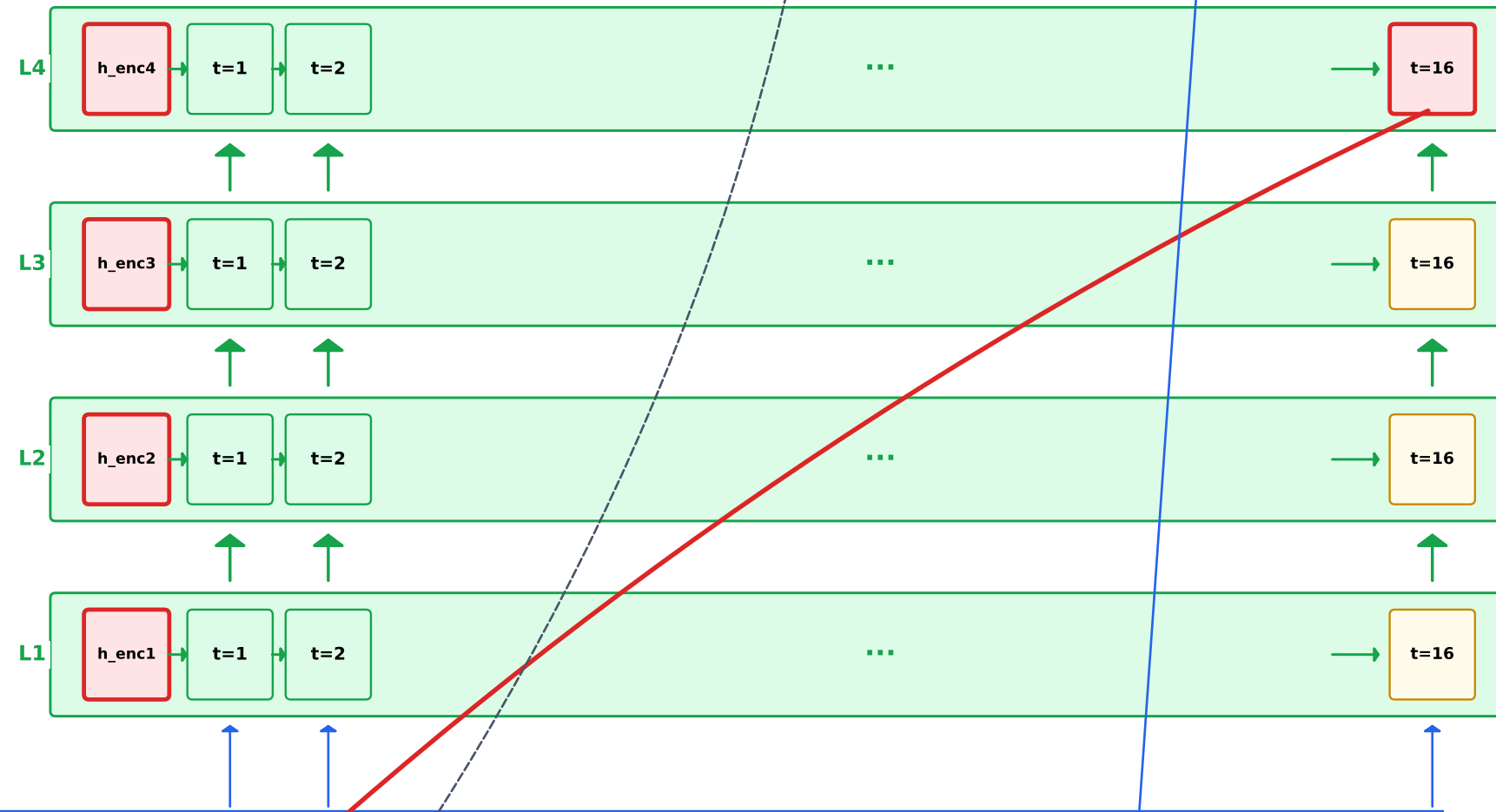
④ Encoder GRU (52 steps)

4 layers × 256 nodes · dropout 0.2

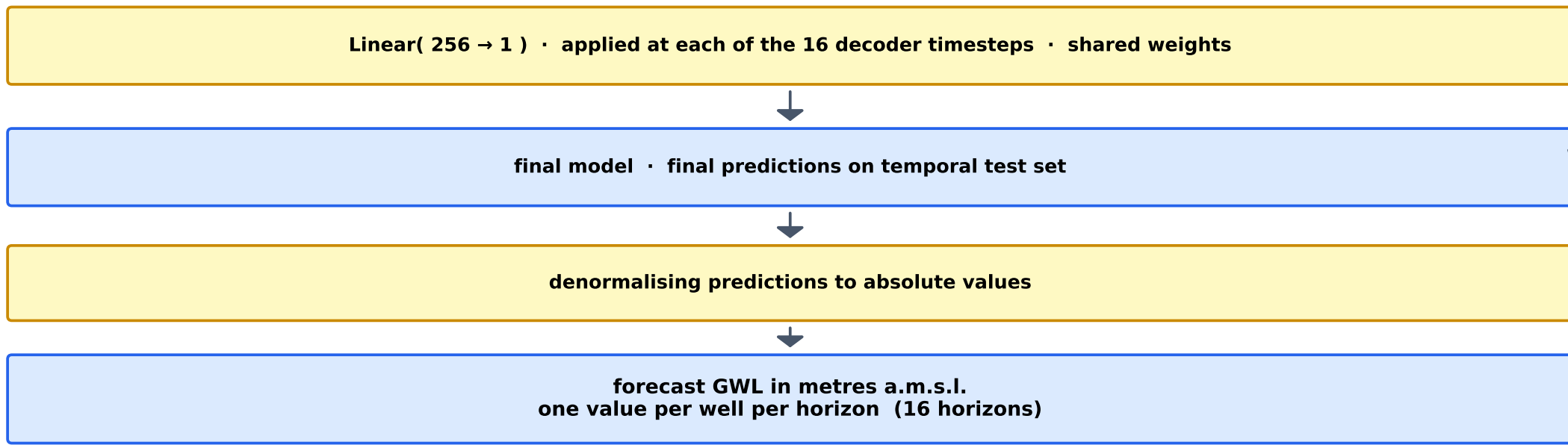


⑤ Decoder GRU (16 steps)

4 layers × 256 nodes · dropout 0.2



⑥ Output Head, Denormalisation & Saved Predictions



MSE Loss
= $\text{mean}((\hat{y} - y_{\text{future}})^2)$
→ backprop → optimizer.step()
→ update weights

⑦ GP Spatial Interpolation

